MIL-M-28787/239A(EC)
3 July 1978
SUPERSEDING
MIL-M-28787/239(EC)
13 January 1976

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DRIVER, NTDS SLOW, DIGITAL, KEY CODE QBQ

This specification is approved for use by Naval Electronics Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic NTDS slow driver module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table II.

Name	Driver, NTDS slow
Family	Digital
Key code	QBQ
Size	1A
Weight	35 grams maximum
Failure rate	0.68 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Supply voltage	
V_{cc}	5.0 ± 0.5 Vdc
V_{ee}	15 ± 2 Vdc
Maximum power dissipation	1.438 W (outputs not loaded)
Data and load input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum
Driver input voltage	
Low level	0.0 V minimum, 0.7 V maximum
High level	3.5 V minimum, 5.5 V maximum
Input current, low level	
Data (V_{IL} = 0 V)	-1.6 mA maximum
Load (V_{IL} = 0 V)	-3.2 mA maximum
Driver (V_{IL} = 0 V)	-5.8 mA maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to. Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Input current, high level	
Data (V_{IH} = 4.0 V)	2.0 µA maximum
(V_{IH} = 5.5 V)	5.0 µA maximum
Load (V_{IH} = 4.0 V)	4.0 µA maximum
(V_{IH} = 5.5 V)	10 µA maximum
Driver (V_{IH} = 4.0 V)	5.0 µA maximum
(V_{IH} = 5.5 V)	15 µA maximum
Output voltage (Q and $\overline{Q}$)	
Low level	0.0 V minimum, 0.4 V maximum
High level	3.8 V minimum, V_{CC} maximum
Output voltage (driver)	
Low level	-10 V minimum, V_{ee} maximum
High level	-1.5 V minimum, 1.5 V maximum
Output drive current maximum capability (Q and $\overline{Q}$)	
Low level (V_{OL} = 0.4 V)	9.6 mA
High level (V_{OH} = 3.8 V)	-60 µA
Output drive current maximum capability (driver)	
Low level	4.0 mA
High level	-16 mA
Propagation delay time (data to Q)	
t_{PLH}	15 ns minimum, 145 ns maximum
t_{PHL}	30 ns minimum, 245 ns maximum
Propagation delay time (data to $\overline{Q}$)	
t_{PLH}	25 ns minimum, 205 ns maximum
t_{PHL}	20 ns minimum, 165 ns maximum
Propagation delay time (driver)	
t_{PLH}	0.84 µs minimum, 4.0 µs maximum
t_{PHL}	0.84 µs minimum, 4.0 µs maximum
Capacitive drive capability (each line driver)	24,000 pF maximum

1.3 Module function. The NTDS slow driver module, designated as key code 08Q, is a Standard Electronic Modules Program (SEMP) electronic module with 0.100 inch (2.54 mm) grid center contacts. This module contains two independent logic holding registers and two independent line driver circuits. Each driver circuit converts 0 to 5 volt logic signals to 0 to -15 volt signals, respectively, and is capable of driving a load of four line receivers, including cabling. The driver output transition rate is less than 5 volts per microsecond and takes less than 6 microseconds to change between the 10 and 90 percent amplitude points. The driver outputs are not short circuit proof.

1.3.1 Input to output relationship. Each holding register has the input to output relationship of table I for positive logic.

TABLE I. Input to output relationship of each register.

Load input	Data input	Q output	$\overline{Q}$ output
L	L	NC	NC
L	H	NC	NC
H	L	L	H
H	H	H	L

NOTES:
1. L = Low-level voltage.
2. H = High-level voltage.
3. NC = No change from previous state.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.68 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be Q8Q.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall hava a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

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3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/239, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed two years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II.

3.9 Power supply voltages. The module shall be capable of conforming to the requirements of table II for power supply voltages (V_{CC}) of 5.0 ±0.5 Vdc and (V_{ee}) of -15 ±2 Vdc, except that the limits for output short circuit current apply only for V_{CC} = 5.5 volts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P01	5
Pulse generator	Item J01	1
Megohm bridge	Item R06	1
DC voltmeter	Item V03	1
DC ammeter	Item X01	1
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
Capacitance meter	Item C05	1
Resistors		
51Ω ± 5%, 1/2 W		2
1.0 kΩ ± 1%, 1/8 W		1
536Ω ± 1%, 1/4 W		4
976Ω ± 1%, 1/2 W		2
3.65 kΩ ± 1%, 1/8 W		1
61.9 kΩ ± 1%, 1/8 W		2
Value as required for figure 10, note 4		2
1.3 kΩ ± 1%, 1/8 W		1
619Ω ± 1%, 1/2 W		1
5.11 kΩ ± 1%, 1/8 W		1
2.49 kΩ ± 1%, 1/8 W		1

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Capacitors	No. required (maximum per test figure)
24,000 pF ± 20%	2
Value as required for figure 10, note 2	2
Value as required for figure 18, note 1(d)	1
Value as required for figure 18, note 2(d)	1
Diodes	
1N4150, or equivalent	2
1N3605, or equivalent	2

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table II.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the SEMP Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modification or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25°± 5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through T shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table II which are marked with an (X) in the test type column shall be performed using the initial limits at 25° ± 5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through T shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table II. Following the return of the module to 25°C, electrical tests D through U shall be performed in the order specified in 4.4.2 using the 25° ± 5°C ambient end-of-life requirements of table II.

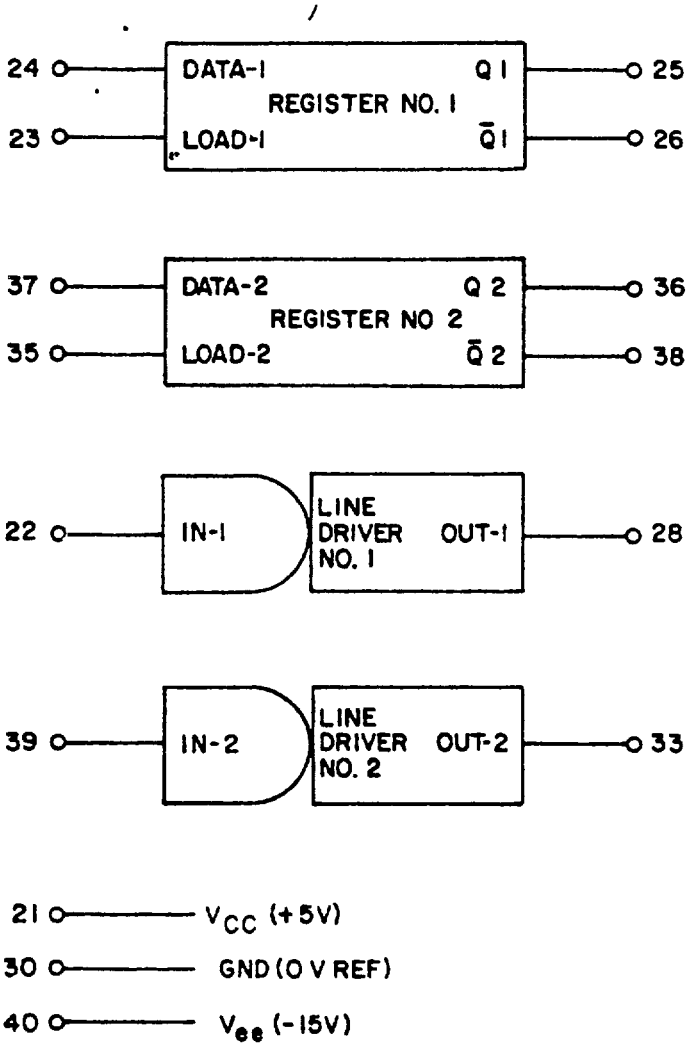
5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

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UNUSED PINS - 1-20, 27, 29, 31, 32, 34.

FIGURE 1. QBQ module function diagram, NTDS slow driver

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TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (See notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Isolation	III	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
B1-B4	V_{out} (low-loaded)	IV	S	0.01	0 8	0.01	0.8	0.01	0.8	0.01	0.8	V
C1, C2	Input breakdown current (load)	V	S	0	10	0	10	0	10	0	10	µA
C3, C4	Input breakdown current (data)	V	S	0	5	0	5	0	5	0	5	µA
C5, C6	Input breakdown current (driver)	V	S	0	15	0	15	0	15	0	15	µA
D1-D4	Output short circuit current	VI	X	-0.7	-2.94	-0.7	-2.94	-0.7	-2.94	-0.7	-2.94	mA
E1-E4	Output voltage, low level (register)	VII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
E5. E6	Output voltage, low level (driver)	VII	X	-10	-17.2	-10	-17.2	-10	-17.2	-10	-17.2	V
F1-F4	Output voltage, high level (register)	VII	X	3.8	4.5	3.8	4.5	3.8	4.5	3.8	4.5	V
F5, F6	Output voltage, high level (driver)	VII	X	-1.5	1.5	-1.5	1.5	-1.5	1.5	-1.5	1.5	V
G1, G2	Output impedance (power off) check	VIII	X	0	175	0	175	0	175	0	175	µA
H1, H2	Input current, high level (data)	IX	X	0.0	2.0	0.0	2.0	0.0	2.0	0.0	2.0	µA
H3, H4	Input current, high level (load)	IX	X	0.0	4.0	0.0	4.0	0.0	4.0	0.0	4.0	µA
H5, H6	Input current, high level (driver)	IX	X	-1,200	1	-1,200	1	-1,200	1	-1,200	1	µA
J1, J2	Input current, low level (data)	X	X	-0.1	-1.6	-0.1	-1.6	-0.1	-1.6	-0.1	-1.6	mA
J3, J4	Input current, low level (load)	X	X	-0.2	-3.2	-0.2	-3.2	-0.2	-3.2	-0.2	-3.2	mA
J5, J6	Input current, low level (driver)	X	X	-0.3	-5.8	-0.3	-5.8	-0.3	-5.8	-0.3	-5.8	mA
K	Power supply current (5 V)	XI	X	5	45	5	45	5	45	5	45	mA
L	Power supply current (-15 V)	XI	X	-1	-70	-1	-70	-1	-70	-1	-70	mA
M1-M4	Output transition time (t_{TLH}), register	XII	S	50	1,000	50	1,000	50	1,000	50	1,000	ns
M5, M6	Output transition time (t_{TLH}), driver	XII	S	2.4	6.0	2.4	6.0	2.4	6.0	2.4	6.0	µs
M7, M8	Output transition time (t_{TLH}), driver	XII	S	1.5	3.75	1.5	3.75	1.5	3.75	1.5	3.75	µs
N1-N4	Output transition time (t_{THL}), register	XII	S	5	75	5	75	5	75	5	75	ns
N5, N6	Output transition time (t_{THL}), driver	XII	S	2.4	6.0	2.4	6.0	2.4	6.0	2.4	6.0	µs

See notes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (See notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N7, N8	Output transition time (t_{THL}), driver	XII	S	1.5	3.75	1.5	3.75	1.5	3.75	1.5	3.75	µs
P1, P2	Propagation delay time (t_{PLH}), data to Q	XII	S	15	140	15	140	15	145	15	145	ns
P3, P4	Propagation delay time (t_{PLH}), data to $\overline{Q}$	XII	S	25	200	25	200	25	205	25	205	ns
P5, P6	Propagation delay time (t_{PLH}), driver	XII	S	0.84	4.0	0.84	4.0	0.84	4.0	0.84	4.0	µs
Q1, Q2	Propagation delay time (t_{PHL}), data to Q	XII	S	30	240	30	240	30	245	30	245	ns
Q3, Q4	Propagation delay time (t_{PHL}), data to $\overline{Q}$	XII	S	20	160	20	160	20	165	20	165	ns
Q5, Q6	Propagation delay time (t_{PHL}), driver	XII	S	0.84	4.0	0.84	4.0	0.84	4.0	0.84	4.0	µs
R1-R4	Driver overshoot and undershoot	XII	Q	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	V
S1-S4	Output crosstalk, high level	XIII	Q	0.0	0.8	0.0	0.8	0.0	0.8	0.0	0.8	V
T1-T4	Output crosstalk, low level	XIII	Q	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
U1, U2	Input capacitance, data	XIV	Q	2.0	15	2.0	15	2.0	15	2.0	15	pF
U3, U4	Input capacitance, load	XIV	Q	4.0	30	4.0	30	4.0	30	4.0	30	pF
U5, U6	Input capacitance, driver	XIV	Q	2.0	25	2.0	25	2.0	25	2.0	25	pF

NOTES:
1. (X) parameters shall be 100 percent tested during module acceptance tests.
2. (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3. (Q) parameters shall be tested during qualification testing only.

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6.2 End-of-life.. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life.

6.3 0 volt reference. The reference for test setups will be 0 volt and not necessarily earth ground.

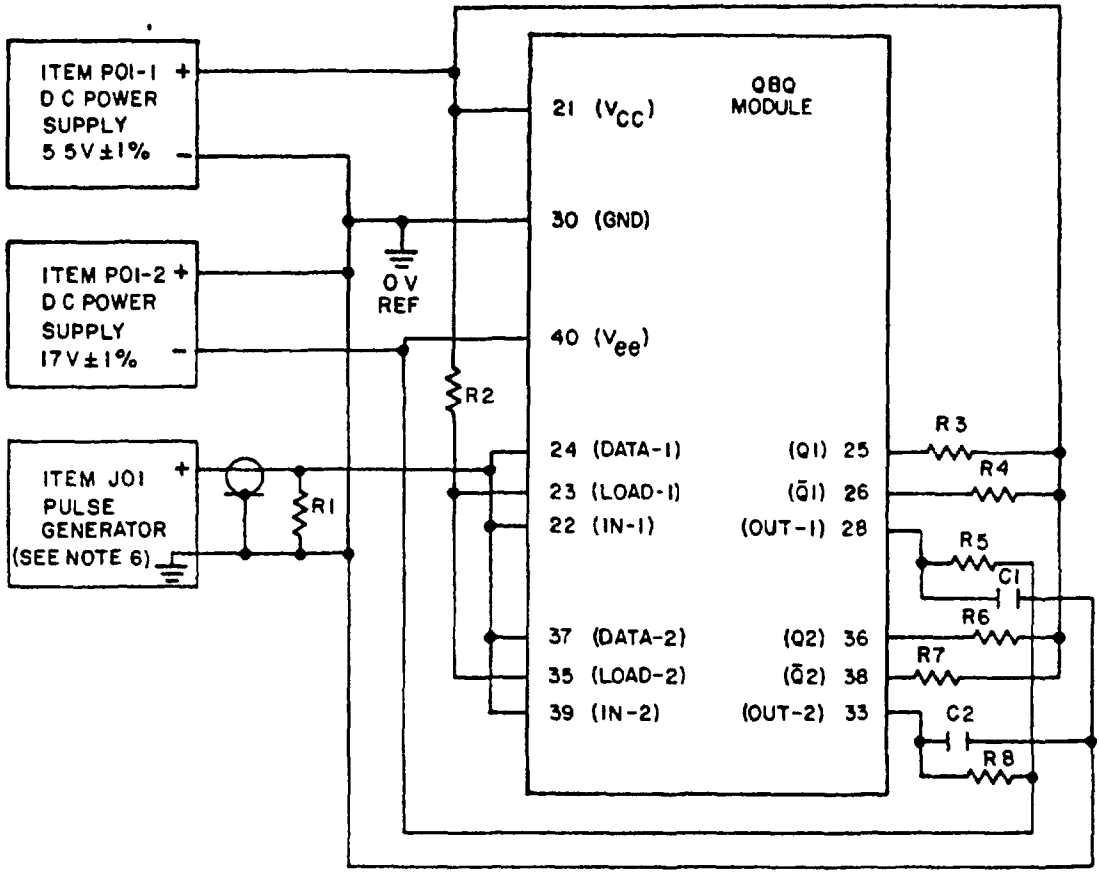
6.4 Decoupling. Decoupling of test voltages and V_{CC} to minimize noise in the test setup is permissible.

Preparing activity:
Navy - EC

Agent:
DLA - ES

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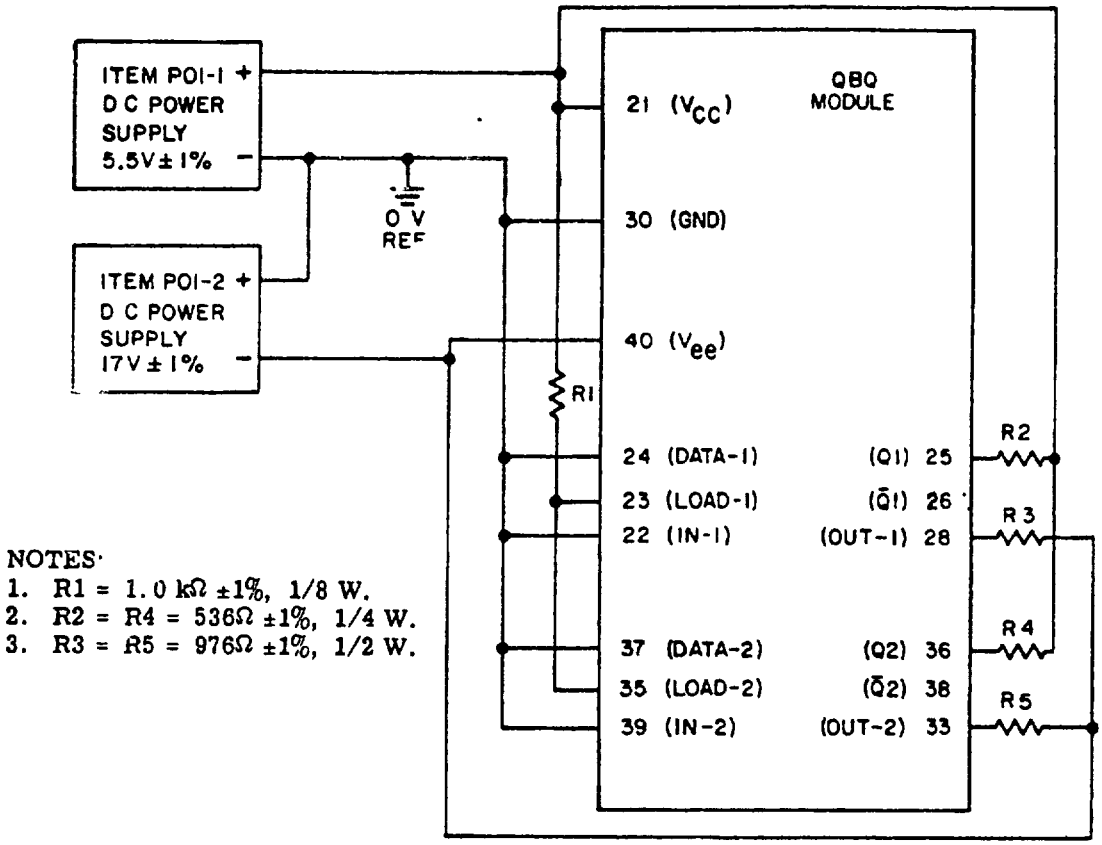
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 1.0 kΩ ±1%, 1/8 W.
3. R3 = R4 = R6 = R7 = 536Ω ±1%, 1/4 W.
4. R5 = R8 = 976Ω ±1%, 1/2 W.
5. C1 = C2 = 24,000 pF ±20%.
6. Pulse generator parameters:
   Amplitude = 4.0 ±0.2 V
   Pulse width = 5.0 ±1.0 µs
   Pulse repetition rate = 100 kHz ±10%.
   Transition time (t_{TLH} and t_{THL}) = 50 ±25 ns.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = 1.0 kΩ ±1%, 1/8 W.
2. R2 = R4 = 536Ω ±1%, 1/4 W.
3. R3 = R5 = 976Ω ±1%, 1/2 W.

FIGURE 3. Component temperature test circuit.

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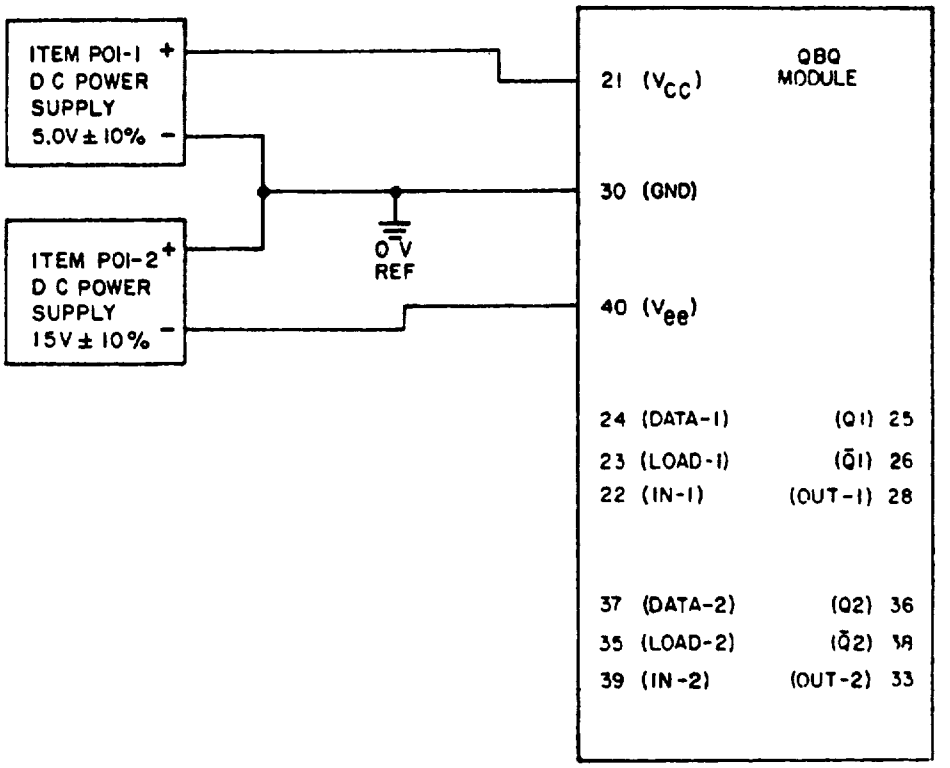


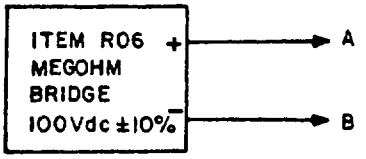
FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° +5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, frame	All pins	Frame	10	-	MΩ	5
A2	Isolation, used to unused pins	1-20,27,29,31, 32,34	All other pins	10	-	MΩ	5

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WARNING·

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

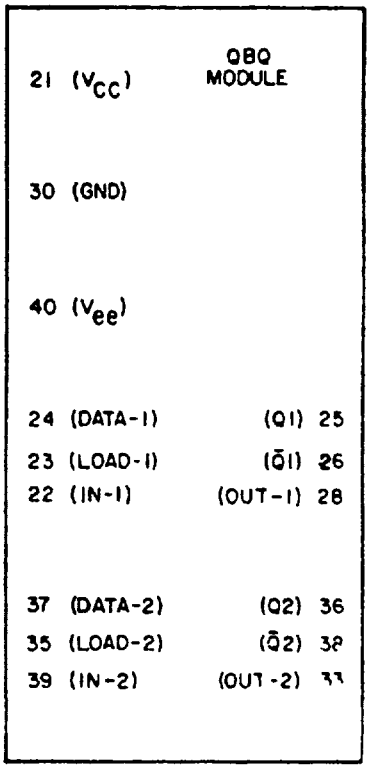


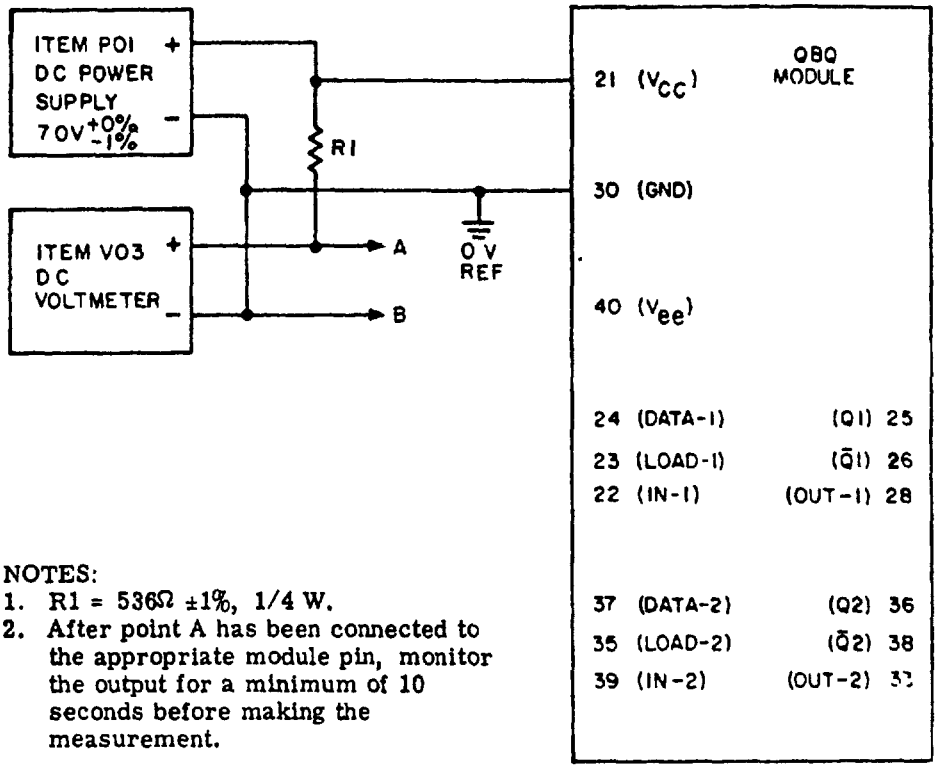
FIGURE 5 Isolation test circuit for table III.

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TABLE IV. V_{out} (low-loaded) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Q1 V_{out} (low-loaded)	25	24	0.01	0.8	V	6
B2	$\overline{Q2}$ ↓	26	Open	↓	↓	↓	↓
B3	Q2 ↓	36	37	↓	↓	↓	↓
B4	$\overline{Q2}$ V_{out} (low-loaded)	38	Open	0.01	0.8	V	6

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NOTES:
1. R1 = 536Ω ±1%, 1/4 W.
2. After point A has been connected to the appropriate module pin, monitor the output for a minimum of 10 seconds before making the measurement.

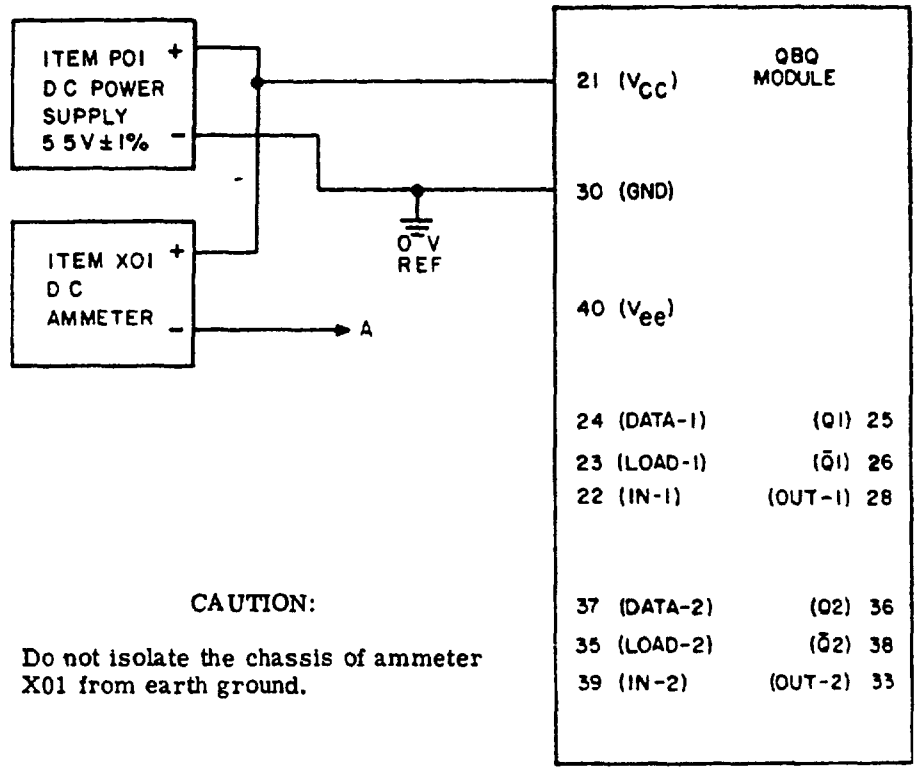
FIGURE 6. V_{out} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
C1	LOAD-1	Input breakdown current	23	0	10	µA	7
C2	LOAD-2		35		10		
C3	DATA-1		24		5		
C4	DATA-2		37		5		
C5	DRIVER IN-1		22		15		
C6	DRIVER IN-2	Input breakdown current	39	0	15	µA	7

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

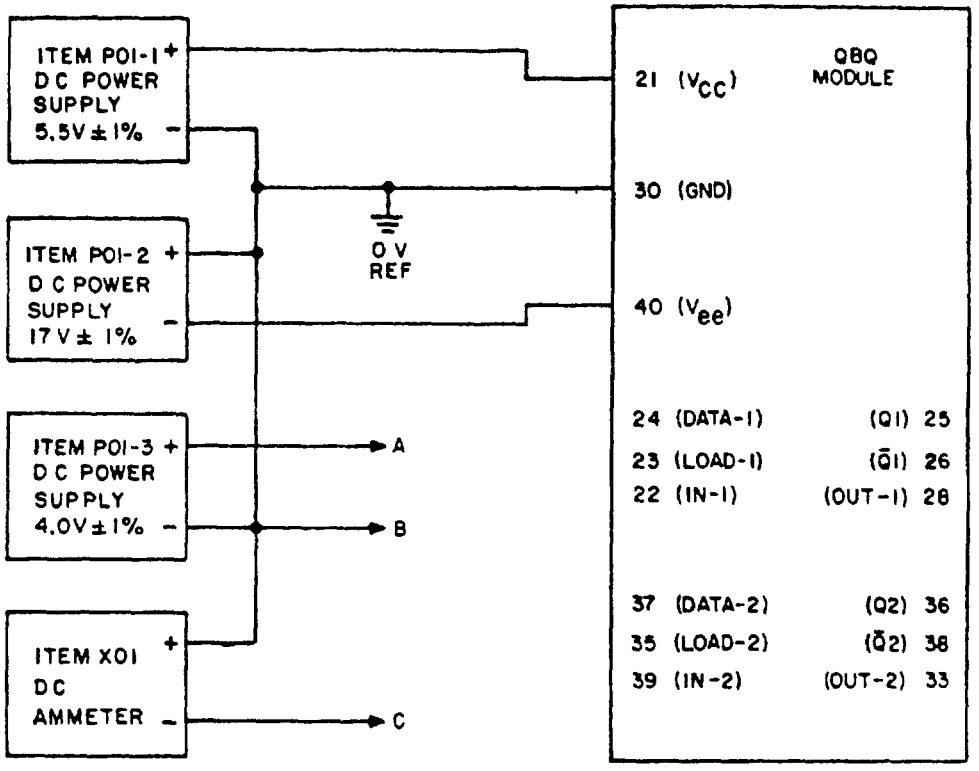
FIGURE 7. Input breakdown current test circuit for table V.

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TABLE VI. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	Q1 Output short circuit current	23, 24	Open	25	-0.7	-2.94	mA	8
D2	$\overline{Q1}$	23	24	26	↓	↓	↓	↓
D3	Q2	35, 37	Open	36	↓	↓	↓	↓
D4	$\overline{Q2}$ Output short circuit current	35	37	38	-0.7	-2.94	mA	8

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: When performing the short circuit current tests, only one output shall be shorted at a time and the measurement taken within 10 seconds.

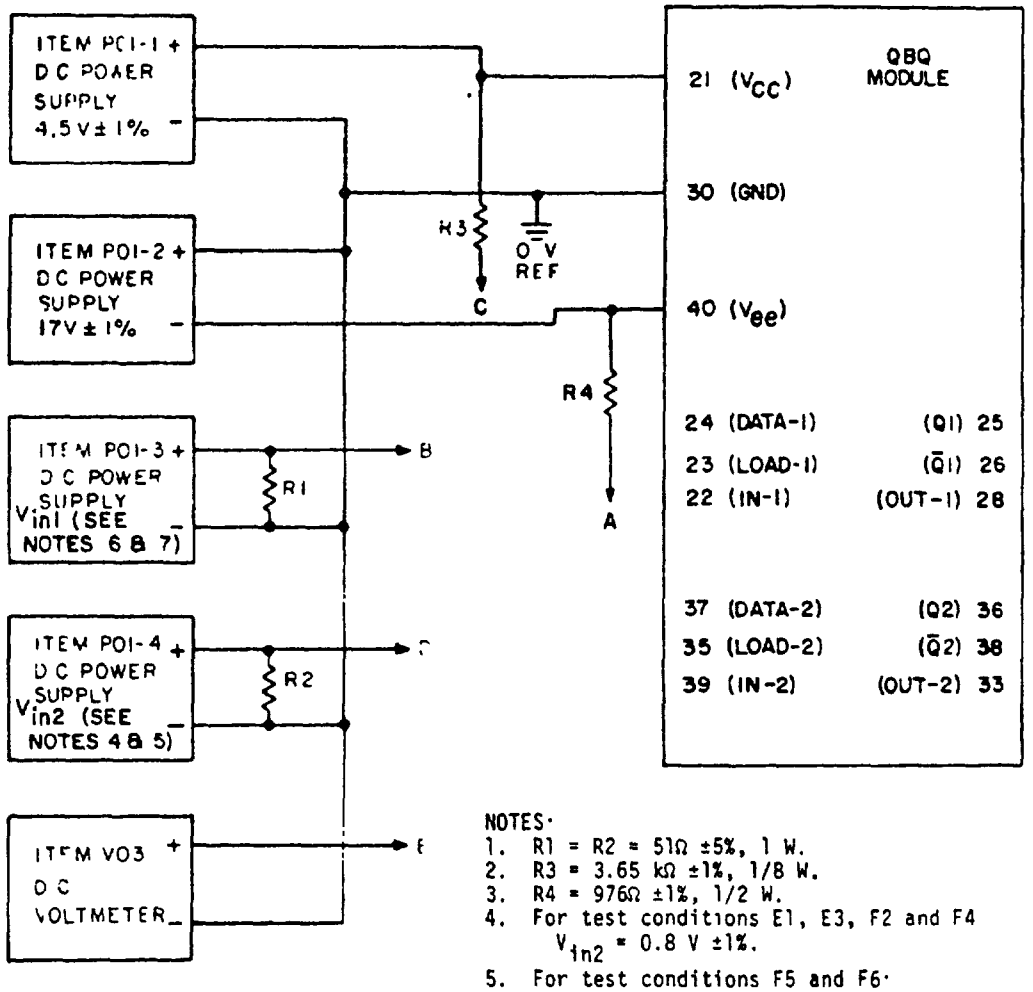
FIGURE 8. Output short circuit current test circuit for table VI.

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TABLE VII. Output voltage (high-level and low-level) test.

Test condition	Parameter		Connect					Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
E1	Q1	Output voltage, low level	Open	23	Open	24	25	0.0	0.4	V	9
E2	$\overline{Q1}$	↓	↓	23,24	↓	Open	26	↓	↓	↓	↓
E3	Q2	↓	↓	35	↓	37	36	↓	↓	↓	↓
E4	$\overline{Q2}$	↓	↓	35,37	Open	Open	38	0.0	0.4	↓	↓
E5	DRIVER OUT-1	↓	↓	22	28	↓	28	-10	-17.2	↓	↓
E6	DRIVER OUT-2	Output voltage, low level	↓	39	33	↓	33	-10	-17.2	↓	↓
F1	Q1	Output voltage, high level	↓	23,24	Open	Open	25	3.8	4.5	↓	↓
F2	$\overline{Q1}$	↓	↓	23	↓	24	26	↓	↓	↓	↓
F3	Q2	↓	↓	35,37	↓	Open	36	↓	↓	↓	↓
F4	$\overline{Q2}$	↓	Open	35	↓	37	38	3.8	4.5	↓	↓
F5	DRIVER OUT-1	↓	28	Open	↓	22	28	-1.5	1.5	↓	↓
F6	DRIVER OUT-2	Output voltage, high level	33	Open	Open	39	33	-1.5	1.5	V	9

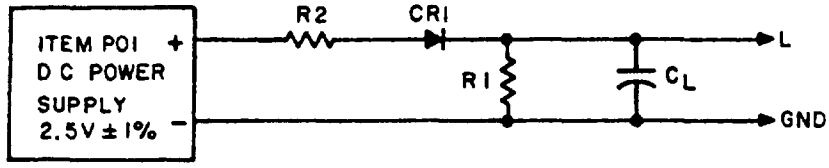
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. R3 = 3.65 kΩ ±1%, 1/8 W.
3. R4 = 976Ω ±1%, 1/2 W.
4. For test conditions E1, E3, F2 and F4 V_{in2} = 0.8 V ±1%.
5. For test conditions F5 and F6: V_{in2} = 0.7 V ±1%.
6. For test conditions E1-E4 and F1-F4: V_{in1} = 2.0 V ±1%
7. For test conditions E5 and E6 V_{in1} = 3.5 V ±1%.
8. Load Q and $\overline{Q}$ outputs of circuit under test with a load as specified on figure 10

FIGURE 9. Output voltage (low-level and high-level) test circuit for table VII

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NOTES:
1. CR1 = 1N4150, or equivalent.
2. C1 = 50 ±5 pF (including probe and parasitic capacitance).
3. R1 = 61.9 kΩ ±1%, 1/8 W.
4. R2 = Select value for a current flow of 9.6 mA ±5% when point L is held at 0.4 volt.
5. C_L is not required for test conditions E and F in table VII.

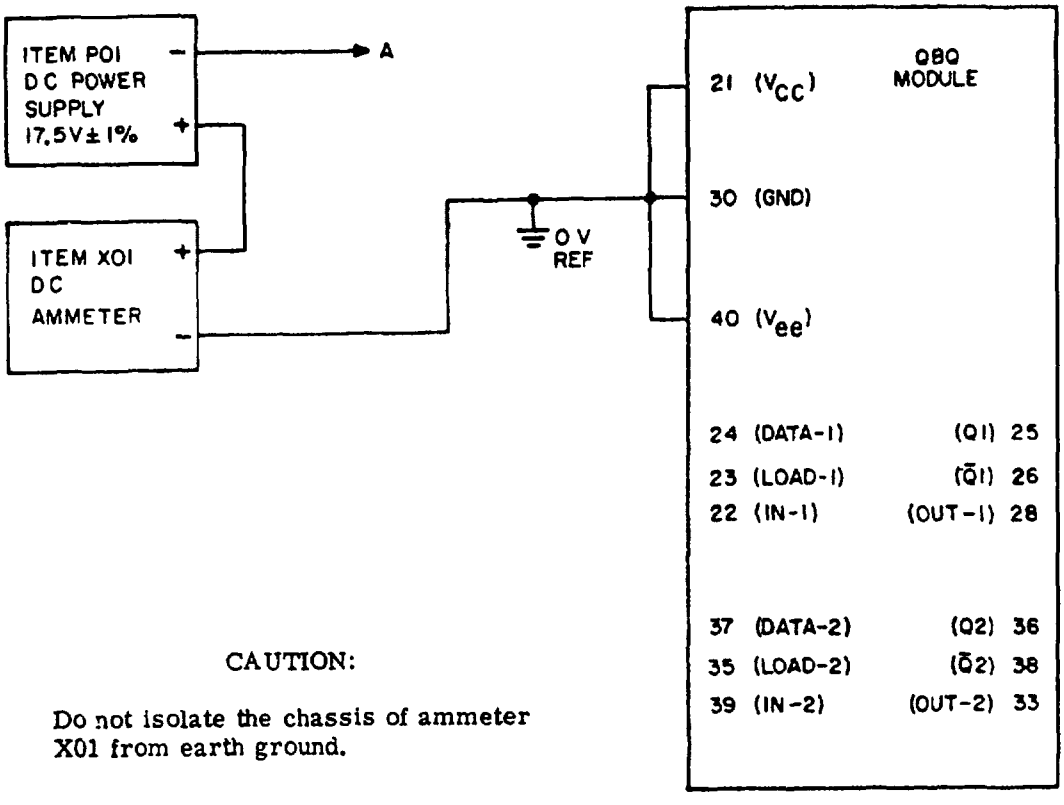
FIGURE 10. QBQ logic load circuit.

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TABLE VIII. Output impedance (power off) check.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
G1	DRIVER OUT-1 Output impedance check	28	0	175	µA	11
G2	DRIVER OUT-2 Output impedance check	33	0	175	µA	11

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CAUTION:

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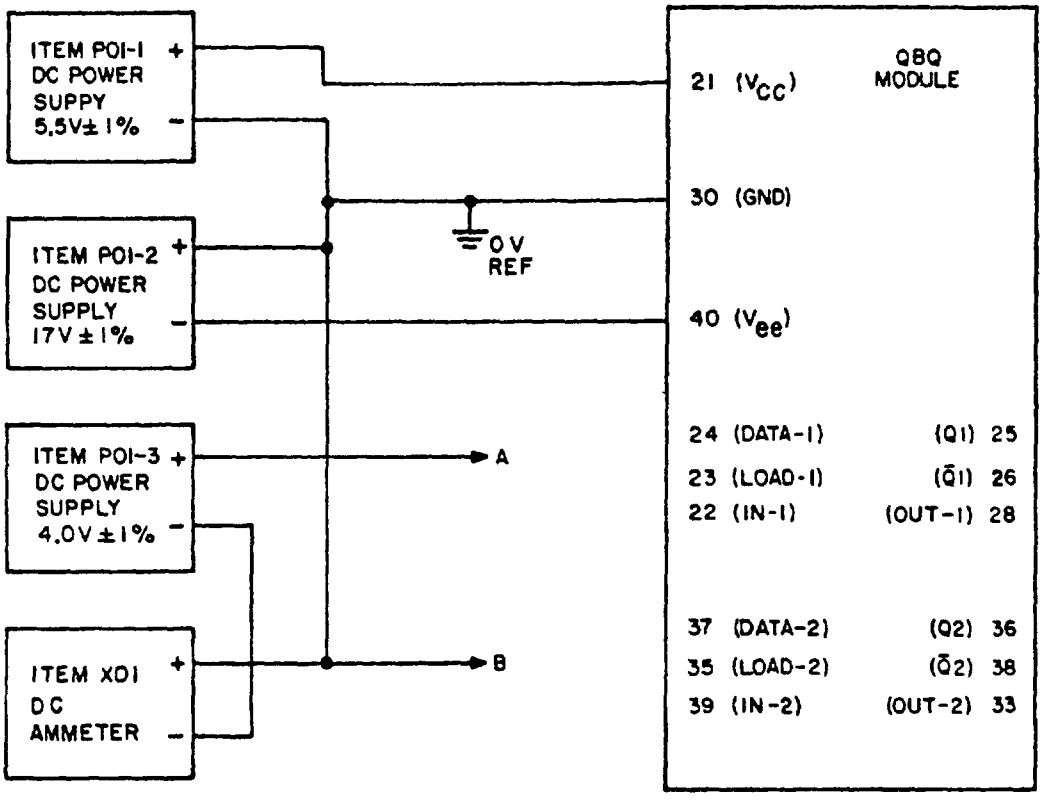
FIGURE 11. Output impedance (power off) check test circuit for table VIII.

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TABLE IX. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	DATA-1 Input current, high level	24	22,23,35,37,39	0.0	2.0	µA	12
H2	DATA-2	37	22,23,35,39	↓	2.0	↓	↓
H3	LOAD-1	23	22,35,39	↓	4.0	↓	↓
H4	LOAD-2	35	22,39	0.0	4.0	↓	↓
H5	DRIVER IN-1	22	39	1,200	1	↓	↓
H6	DRIVER IN-2 Input current, high level	39	Open	-1,200	1	µA	12

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CAUTION:

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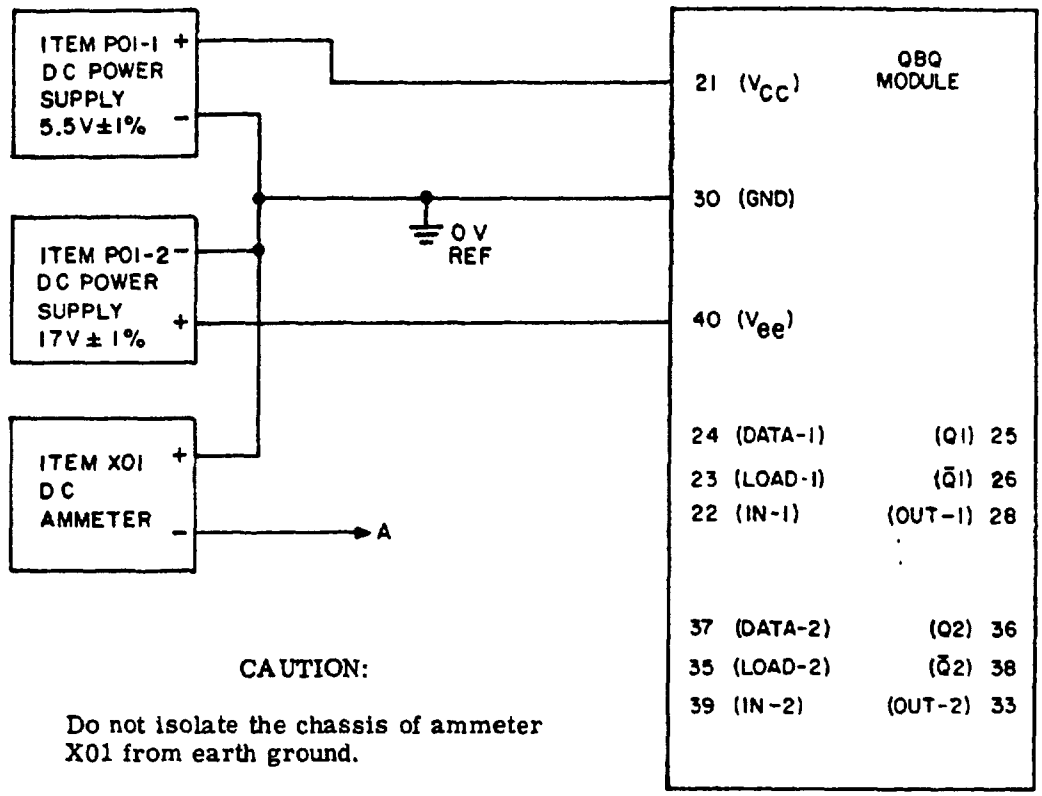
FIGURE 12. Input current (high-level) test circuit for table IX.

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TABLE X. Input current (low-level) tests.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C Ambient Min	Initial limits 25°C ±5°C Ambient Max	Unit	Test figure
J1	DATA-1 Input current, low level	24	-0.1	-1.6	mA	13
J2	DATA-2	37	-0.1	-1.6		
J3	LOAD-1	23	-0.2	-3.2		
J4	LOAD-2	35	-0.2	-3.2		
J5	DRIVER IN-1	22	-0.3	-5.8		
J6	DRIVER IN-2 Input current, low level	39	-0.3	-5.8	mA	13

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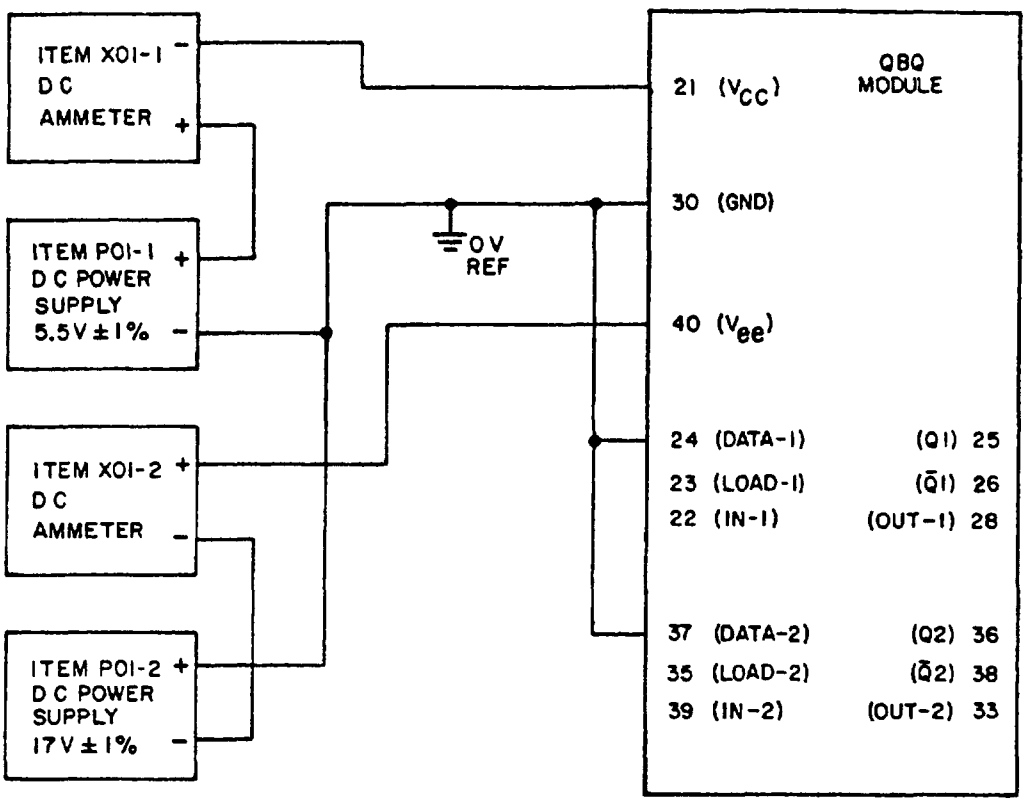
FIGURE 13. Input current (low-level) test circuit for table X.

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TABLE XI. Power supply current test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
K	Power supply current (5 V)	5	45	mA	14
L	Power supply current (-15 V)	-1	-70	mA	14

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NOTE Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.

FIGURE 14. Power supply current test circuit for table XI.

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TABLE XII. Output transition time, propagation delay time, and driver overshoot and undershoot test.

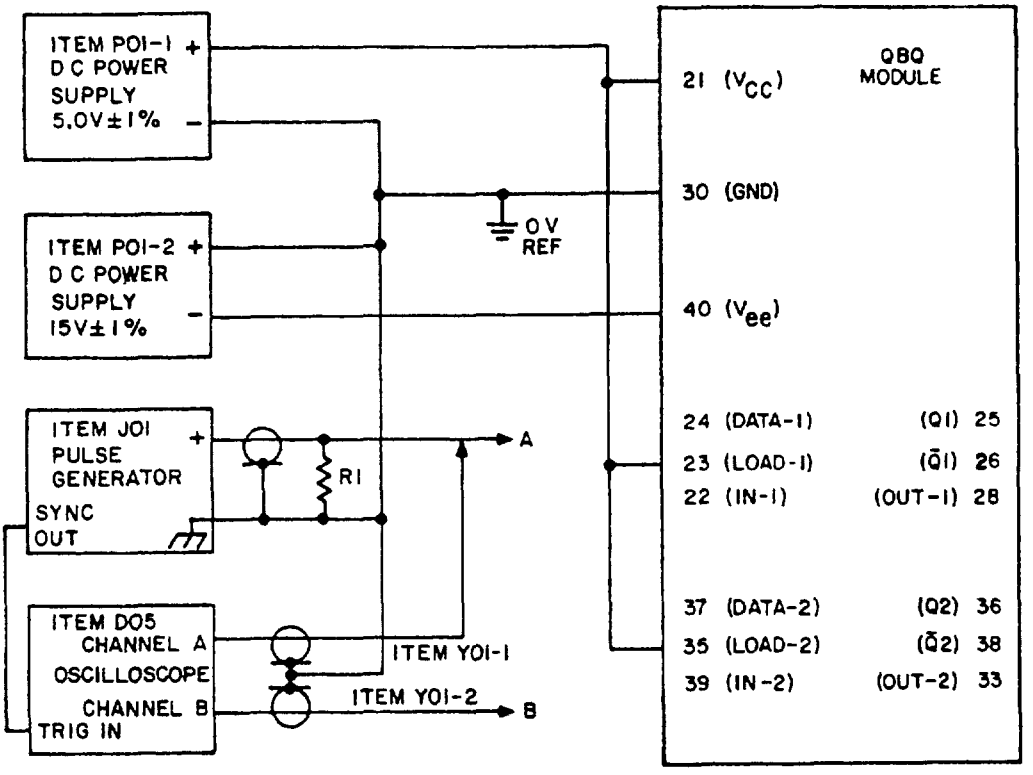
Test condition	Parameter		Connect		Initial limits 25° ± 5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
M1	Q1	Output transition time (t_{TLH})	24	25	50	1,000	ns	15, 16
M2	$\overline{Q1}$	↓	24	26	↓	↓	↓	↓
M3	Q2	↓	37	36	↓	↓	↓	↓
M4	$\overline{Q2}$	Output transition time (t_{TLH})	37	38	50	1,000	ns	15, 16
M5	DRIVER OUT-1	Output transition time (t_{TLH}), C_L = 24,000 pF	22	28	2.4	6.0	µs	15, 17
M6	DRIVER OUT-2	Output transition time (t_{TLH}), C_L = 24,000 pF	39	33	2.4	6.0	↓	↓
M7	DRIVER OUT-1	Output transition time (t_{TLH}), C_L = 100 pF	22	28	1.5	3.75	↓	↓
M8	DRIVER OUT-2	Output transition time (t_{TLH}), C_L = 100 pF	39	33	1.5	3.75	µs	15, 17
N1	Q1	Output transition time (t_{THL})	24	25	5	75	ns	15, 16
N2	$\overline{Q1}$	↓	24	26	↓	↓	↓	↓
N3	Q2	↓	37	36	↓	↓	↓	↓
N4	$\overline{Q2}$	Output transition time (t_{THL})	37	38	5	75	ns	15, 16
N5	DRIVER OUT-1	Output transition time (t_{THL}), C_L = 24,000 pF	22	28	2.4	6.0	µs	15, 17
N6	DRIVER OUT-2	Output transition time (t_{THL}), C_L = 24,000 pF	39	33	2.4	6.0	↓	↓
N7	DRIVER OUT-1	Output transition time (t_{THL}), C_L = 100 pF	22	28	1.5	3.75	↓	↓
N8	DRIVER OUT-2	Output transition time (t_{THL}), C_L = 100 pF	39	33	1.5	3.75	µs	15, 17
P1	DATA-1 to Q1	Propagation delay time (t_{PLH})	24	25	15	140	ns	15, 16
P2	DATA-2 to Q2	Propagation delay time (t_{PLH})	37	36	15	140	ns	15, 16

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TABLE XII. Output transition time, propagation delay time, and driver overshoot and undershoot test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ± 5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
P3	DATA-1 to $\overline{Q1}$	Propagation delay time (t_{PLH})	24	26	25	200	ns	15, 16
P4	DATA-2 to $\overline{Q2}$		37	38	25	200	ns	15, 16
P5	DRIVER IN-1 to OUT-1		22	28	0.84	4.0	µs	15, 17
P6	DRIVER IN-2 to OUT-2	Propagation delay time (t_{PLH})	39	33	0.84	4.0	µs	15, 17
Q1	DATA-1 to Q1	Propagation delay time (t_{PHL})	24	25	30	240	ns	15, 16
Q2	DATA-2 to Q2		37	36	30	240		
Q3	DATA-1 to $\overline{Q1}$		24	26	20	160		
Q4	DATA-2 to $\overline{Q2}$		37	38	20	160	ns	15, 16
Q5	DRIVER IN-1 to OUT-1		22	28	0.84	4.0	µs	15, 17
Q6	DRIVER IN-2 to OUT-2	Propagation delay time (t_{PHL})	39	33	0.84	4.0	µs	15, 17
R1	DRIVER OUT-1	Driver overshoot	22	28	0.0	1.0	V	15, 17
R2	DRIVER OUT-1	Driver undershoot	22	28				
R3	DRIVER OUT-2	Driver overshoot	39	33				
R4	DRIVER OUT-2	Driver undershoot	39	33	0.0	1.0	V	15, 17

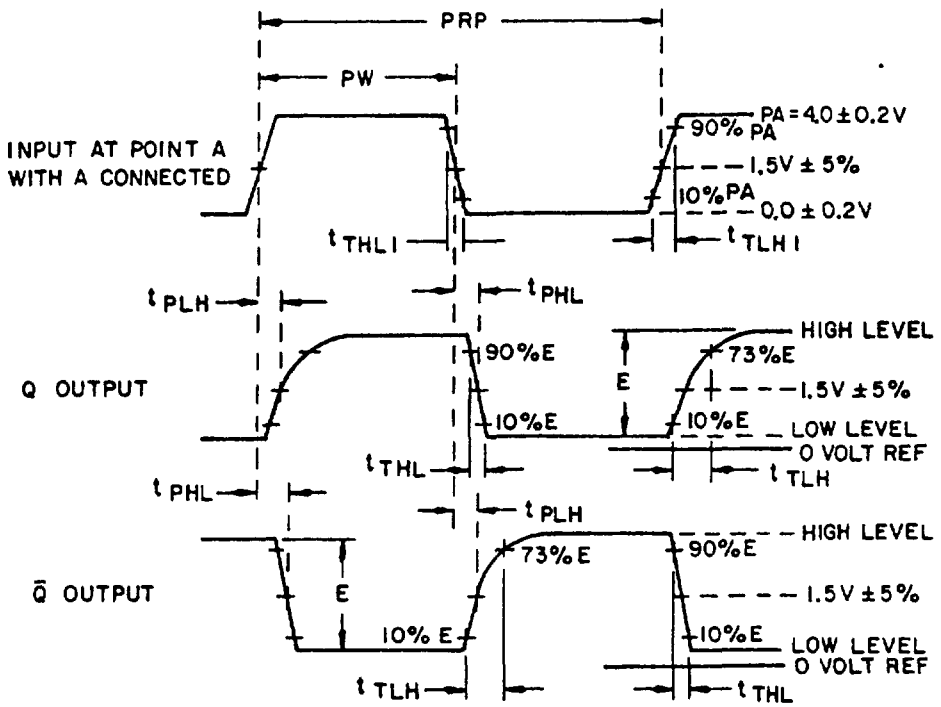
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load Q and $\bar{Q}$ outputs of circuit under test with load as specified on figure 10.
3. Load driver output of circuit under test with load as specified on figure 18.

FIGURE 15. Output transition time, propagation delay time, and driver overshoot and undershoot test circuit for table XII.

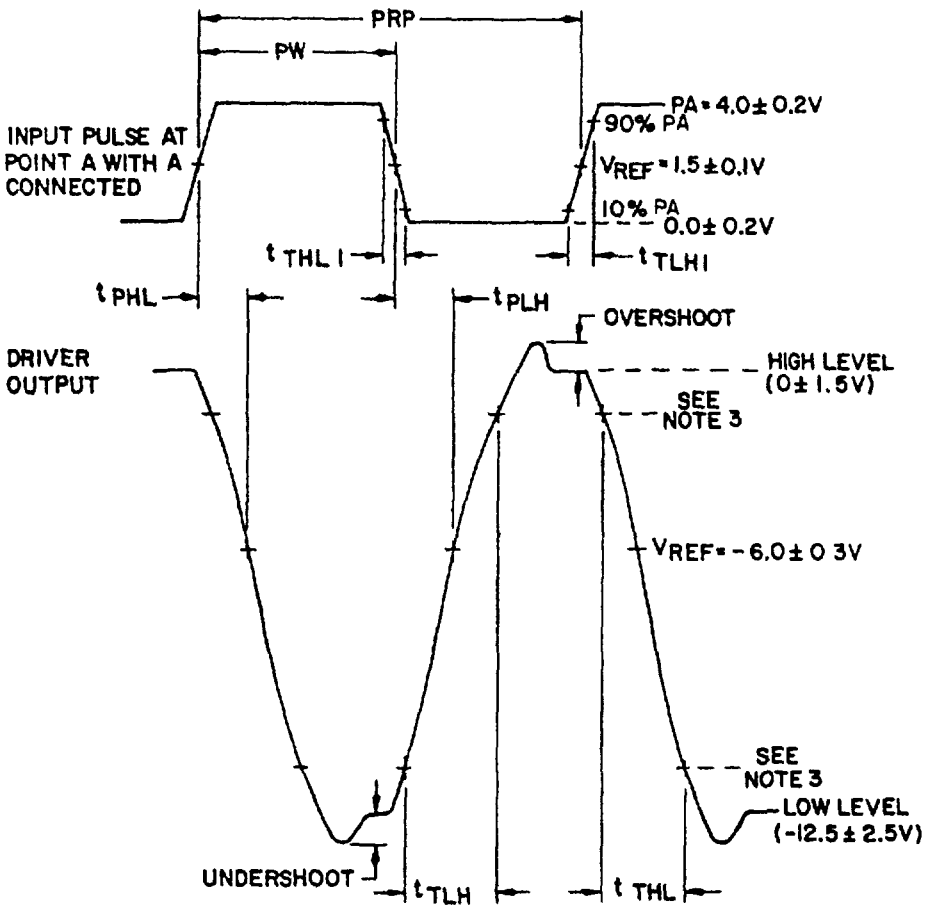
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NOTES:
1. Input pulse parameters:
   PW = 10 ±0.5 μs
   PRP = 20.0 ±1.0 μs
   $t_{TLH1} = t_{THL1}$ = 20 ±2 ns.
2. Before making measurements, verify that for Q and $\bar{Q}$ outputs the high level is 3.8 volts minimum and the low level is 0.4 volt maximum, for the driver output, the high level is -1.5 volts minimum and the low level is -12 volts maximum.
3. All voltage levels are referenced to ground.

FIGURE 16. Output transition time and propagation delay time test circuit waveforms for table XII.

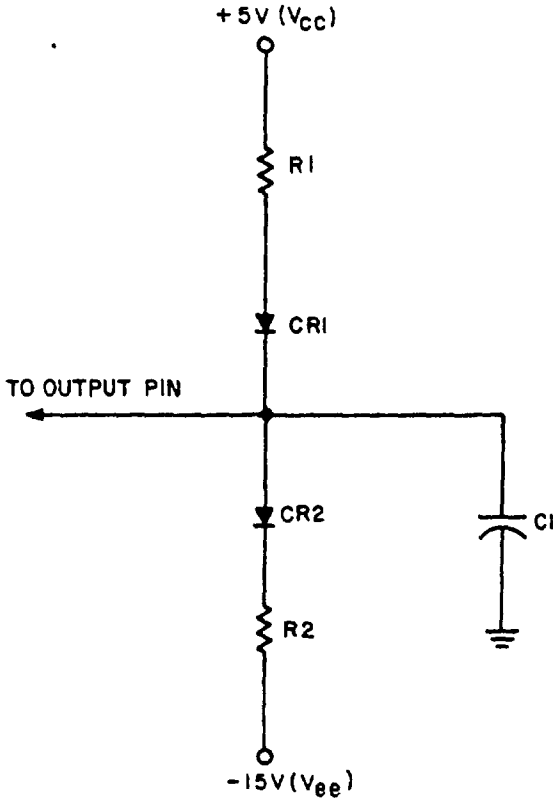
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NOTES:
1. Input pulse parameters.
   PW = 20 ±1 μs
   PRP = 30 ±3 μs
   $t_{TLH1} = t_{THL1} = 20 \pm 2$ ns.
2. Before making measurements, verify that the high level is 0 ±1.5 V and the low level is -12.5 ±2.5 V.
3. For test conditions M5, M6, N5, and N6, t_{TLH} and t_{THL} are measured between the 10% and 90% points. For test conditions M7, M8, N7, and N8, t_{TLH} and t_{THL} are measured between the 25% and 75% points.

FIGURE 17. Output transition time (driver), propagation delay time (driver), and driver overshoot and undershoot test circuit waveforms for table XII.

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NOTES:
1. For test conditions M5, M6, N5, N6, P5, P6, and R1-R4:

   (a) CR1 = CR2 = 1N3605, or equivalent.
   (b) R1 = 1.3 kΩ ±1%, 1/8 W.
   (c) R2 = 619Ω ±1%, 1/2 W.
   (d) C1 = 24,000 pF ±10% (including probe and parasitic capacitance).

2. For test conditions M7, M8, N7, and N8:

   (a) CR1 = CR2 = 1N3605, or equivalent.
   (b) R1 = 5.11 kΩ ±1%, 1/8 W.
   (c) R2 = 2.49 kΩ ±1%, 1/8 W.
   (d) C1 = 100 pF ±10% (including probe and parasitic capacitance).

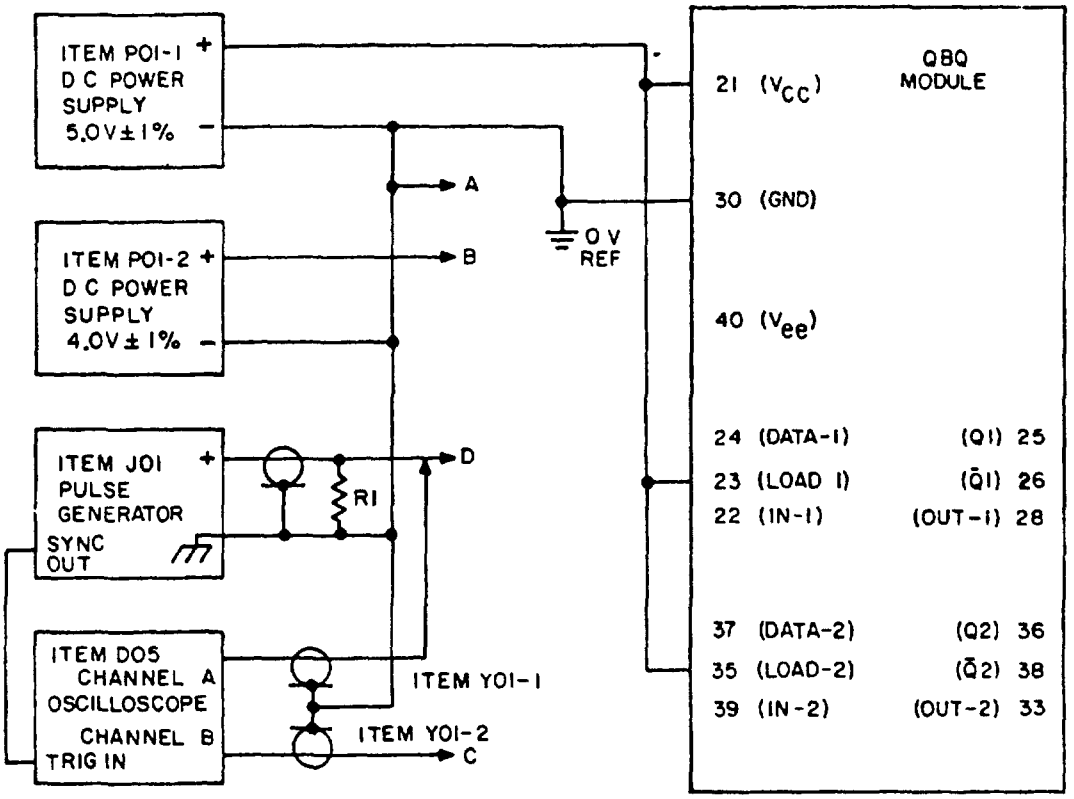
FIGURE 18. Driver output load circuit.

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TABLE XIII. Output crosstalk (high-level and low-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	Q1 Output crosstalk, high level	Open	24	25	22, 37,39	0.0	0.8	V	19, 20
S2	$\overline{Q1}$	24	Open	26	22, 37,39				
S3	Q2	Open	37	36	22, 24,39				
S4	$\overline{Q2}$ Output crosstalk, high level	37	Open	38	22, 24,39		0.8		
T1	Q1 Output crosstalk, low level	24	Open	25	37		0.4		
T2	$\overline{Q1}$	Open	24	26	37				
T3	Q2	37	Open	36	24				
T4	$\overline{Q2}$ Output crosstalk, low level	Open	37	38	24	0.0	0.4	V	19, 20

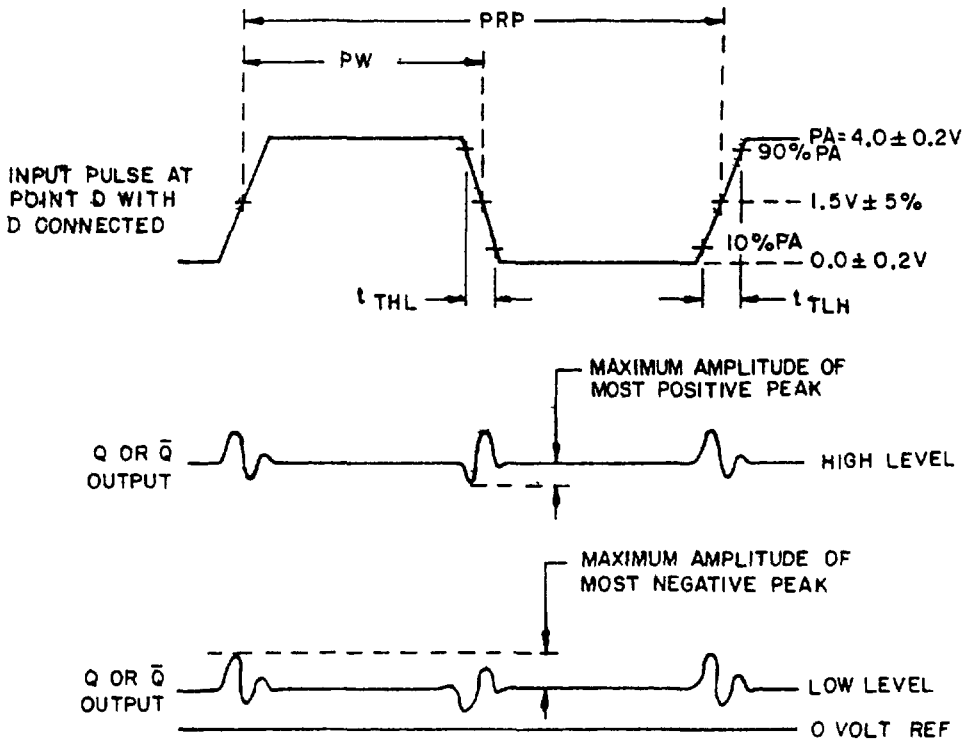
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load all Q and $\overline{Q}$ outputs with the load as specified on figure 10.

FIGURE 19. Output crosstalk (high-level and low-level) test circuit for table XIII.

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NOTE: Input pulse parameters:
PW = 6.5 ±0.5 μs
t_{TLH} = t_{THL} = 20 ±2 ns
PRP = 13.0 ±1.0 μs.

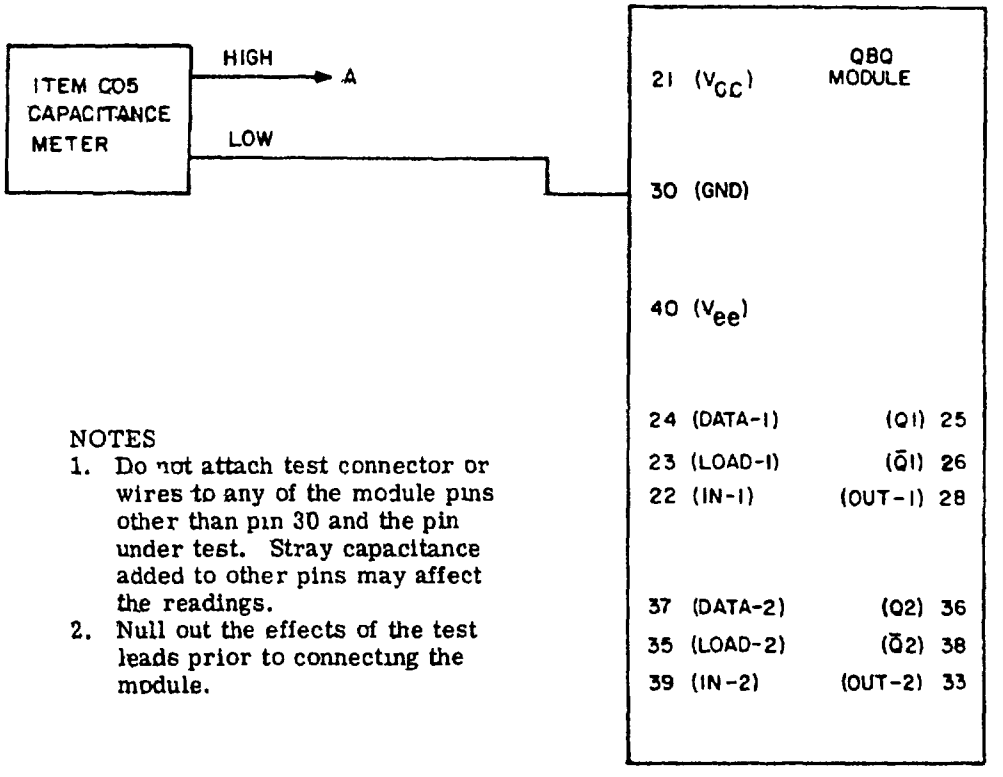
FIGURE 20. Output crosstalk (high-level and low-level) test circuit waveforms for table XIII.

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TABLE XIV. Input capacitance test.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
U1	DATA-1	Input capacitance	24	2.0	15	pF	21
U2	DATA-2	↓	37	2.0	15	↓	↓
U3	LOAD-1	↓	23	4.0	30	↓	↓
U4	LOAD-2	↓	35	4.0	30	↓	↓
U5	DRIVER IN-1	↓	22	2.0	25	↓	↓
U6	DRIVER IN-2	Input capacitance	39	2.0	25	pF	21

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NOTES
1. Do not attach test connector or wires to any of the module pins other than pin 30 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

FIGURE 21 Input capacitance test circuit for table XIV.

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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